20210710 高等数学一（II）期末考试试题回忆

这该死的，说好的降低难度呢呜呜
顾老师我爱你，小斌也是）

1.试确定α(α>0)的范围，使得函数$f(x,y)=\begin{cases}(x^2+y^2)^{\alpha}\sin\left(\dfrac{1}{x^2+y^2}\right) & x^2+y^2\neq 0\\ 0 & x^2+y^2=0\end{cases}$

在（0,0）处可微.

2.计算第二型曲线积分$\oint_L(\sin y+xy^2)dy-(\cos x+x^2y)dx$，其中 L 为闭曲线$x^2+y^2=4$的逆时针方向边界.

3.设三角形的周长为 2p，求怎样的三角形绕其中一条边旋转的得到的旋转体的体积最大.

感谢广宇，梅开二度（可恶！以后必做课本全部课后习题） ///(^v^)\\\

习 题 6.9

4. 已知三角形的周长为 $2p$，问怎样的三角形绕自己的一边旋转一周所得的体积最大.

4.求解微分方程 $y\ln ydx+(x-\ln y)dy=0$.

5.求解微分方程 $\dfrac{d^2y}{dx^2}=\left(\dfrac{dy}{dx}\right)^3+\dfrac{dy}{dx}$.

6.求解二阶常微分方程 $y''+y=e^x+\cos x$ 的通解.

7.判断级数$\dfrac{1}{\sqrt{2}-1}-\dfrac{1}{\sqrt{2}+1}+\dfrac{1}{\sqrt{3}-1}-\dfrac{1}{\sqrt{3}+1}+\dfrac{1}{\sqrt{4}-1}-\dfrac{1}{\sqrt{4}+1}+\ldots$的散敛性.

8.试讨论α(α>0)的取值，判断级数$\sum_{n=1}^{\infty}\left(\dfrac{1}{n}-\sin\dfrac{1}{n}\right)^{\alpha}$的散敛性.

9.判断函数项级数$\sum_{n=1}^{\infty}2^n\sin\dfrac{x}{3^n}$在下列区间的一致收敛性.

(1) $[-M,M]$ 其中M为正的常数

(2) $[-\infty,\infty]$

10.求幂级数$\sum_{n=1}^{\infty}\dfrac{(2n)!}{(n!)^2}x^{4n}$的收敛半径$R$.

11.求幂级数$\sum_{n=1}^{\infty}\dfrac{x^{4n+1}}{4n+1}$的和函数.

12.将函数$f(x)=\ln\left(\sqrt{x^2+1}-x\right)$在$x=0$处的幂级数展开式.

by cyc

T1 $\Delta f(x,y) = f(0+\Delta x, 0+\Delta y) - f(0,0) = (\Delta x^2+\Delta y^2)^{\alpha} \cdot \sin\left(\frac{1}{\Delta x^2+\Delta y^2}\right)$

而 $\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = \lim\limits_{\Delta x\to 0} \frac{f(0+\Delta x,0)-f(0,0)}{\Delta x} = \lim\limits_{\Delta x\to 0} \frac{\Delta x^{2\alpha}\cdot \sin\frac{1}{\Delta x^2} - 0}{\Delta x}$

$= \lim\limits_{\Delta x\to 0} (\Delta x^{2\alpha-1}) \cdot \left(\sin\frac{1}{\Delta x^2}\right)$

若 $f(x,y)$ 可微，则其偏导存在，$2\alpha-1>0$，$\alpha>\frac{1}{2}$

此时 $\lim\limits_{\Delta x\to 0} \Delta x^{2\alpha-1} = 0$，$\sin\frac{1}{\Delta x^2}$ 有界，$\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = 0$

对 $\left.\frac{\partial f}{\partial y}\right|_{(0,0)}$ 同理

又：$\lim\limits_{\rho\to 0} \frac{\Delta f - df}{\rho} = \lim\limits_{\rho\to 0} \frac{\Delta f}{\rho} = \lim\limits_{\rho\to 0} \rho^{2\alpha-1} \sin\left(\frac{1}{\rho^2}\right)$

$\lim\limits_{\rho\to 0} \rho^{2\alpha-1} = 0$，$\sin\frac{1}{\rho^2}$ 有界，故 $\lim\limits_{\rho\to 0} \frac{\Delta f - df}{\rho} = 0$

可微恒成立

故 $\alpha>\frac{1}{2}$

T2 记 $P = -(\cos x + x^2 y)$　$Q = (\sin y + xy^2)$

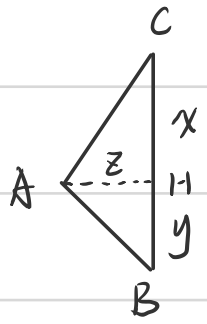
$\frac{\partial P}{\partial y} = -x^2$　$\frac{\partial Q}{\partial x} = y^2$　偏导连续

对闭曲线 L，由格林公式

原式 $= \iint\limits_D y^2 + x^2 \, dx dy$　$D: x^2+y^2 \le 4$

$= 4\int_0^{\frac{\pi}{2}} d\theta \int_0^2 r^3 \, dr = 8\pi$

T3



作BC边上高H，令 CH=x，BH=y，AH=z

有 $x+y+\sqrt{x^2+z^2}+\sqrt{x^2+y^2}=2p$

不妨绕BC旋转，

有 $V=\frac{1}{3}(x+y)\cdot\pi z^2$ 其实可以求 $F=(x+y)z^2$

记 $F(x,y,z,\lambda)=\frac{\pi}{3}(x+y)z^2-\lambda(x+y+\sqrt{x^2+z^2}+\sqrt{y^2+z^2}-2p)$

① $\frac{\partial F}{\partial x}=\frac{\pi}{3}z^2-\lambda-\frac{\lambda x}{\sqrt{x^2+z^2}}=0$

② $\frac{\partial F}{\partial y}=\frac{\pi}{3}z^2-\lambda-\frac{\lambda y}{\sqrt{y^2+z^2}}=0$

③ $\frac{\partial F}{\partial z}=\frac{2\pi}{3}z(x+y)-\frac{\lambda z}{\sqrt{x^2+z^2}}=0$

④ $\frac{\partial F}{\partial \lambda}=x+y+\sqrt{x^2+z^2}+\sqrt{y^2+z^2}-2p=0$

由①,② $\frac{\lambda x}{\sqrt{x^2+z^2}}=\frac{\lambda y}{\sqrt{y^2+z^2}}\Rightarrow x=y$ ($\lambda\neq0$，否则 $z=0$)

代入④ $x+\sqrt{x^2+z^2}=p\Rightarrow x^2+z^2=(p-x)^2\Rightarrow z^2=p^2-2px$

代入③ $\frac{4}{3}\pi xz-\frac{\lambda z}{p-x}=0\Rightarrow\lambda=\frac{4}{3}\pi x(p-x)$

代入① $\frac{\pi}{3}z^2-\frac{4}{3}\pi x(p-x)\left(1+\frac{x}{p-x}\right)=0$

解得 $x=\frac{p}{2}$ 或 $\frac{p}{4}$；$x=\frac{p}{2}$ 时，$x+y=p$，$AC+AB=2p-p=p$

不构成三角形

故 $x=y=\frac{p}{4}$，即 $AC=AB=\frac{3}{4}p$，$BC=\frac{p}{2}$ 的三角形符合

需要说明 max？

T_4 $y\ln y\,dx + (x-\ln y)\,dy = 0 \quad y>0$

$$\Rightarrow \frac{dx}{dy} = \frac{\ln y - x}{y\ln y} = \frac{1}{y} - \frac{x}{y\ln y} \quad (y\neq 1)$$

$$\Rightarrow \frac{dx}{dy} + \frac{1}{y\ln y}x = \frac{1}{y}$$

考虑 $\frac{dx}{dy} + \frac{x}{y\ln y} = 0$, $\frac{dx}{x} = -\frac{dy}{y\ln y}$ $(x\neq 0)$

$$\ln x + \ln C = -\ln(\ln y)$$

$$Cx = \frac{1}{\ln y}$$

使用常数变易法 $u(y)\cdot x = \frac{1}{\ln y}$, $x\frac{du}{dy} + u\frac{dx}{dy} = -\frac{1}{y\ln^2 y}$ 求导!

代入得 $\frac{1}{u}\left(-\frac{1}{y\ln^2 y} - \frac{x\,du}{dy}\right) + \frac{1}{u\ln^2 y\, y} = \frac{1}{y}$

即 $-\frac{x\,du}{u\,dy} = \frac{1}{y} \Rightarrow -\frac{du}{u^2} = \frac{\ln y}{y}dy$

$$\Rightarrow \frac{1}{u} = \frac{1}{2}(\ln^2 y) + C$$

$$\Rightarrow x = \frac{1}{2}\ln y + \frac{C}{\ln y}$$

$y=1$ 时 方程恒成立

$x=0$ 时 从属于 $C=0$ 时 $y=1$ 的情况 ?

其实 $x = \frac{C}{\ln y} \Rightarrow x = \frac{u(y)}{\ln y}$ $\frac{dx}{dy} = \frac{\frac{du}{dy}\ln y - \frac{u}{y}}{\ln^2 y}$

代入 $\frac{du}{\ln y\,dy} - \frac{u}{y\ln^2 y} + \frac{u}{y\ln^2 y} = \frac{1}{y}$

$\Rightarrow du = \frac{\ln y}{y}dy \Rightarrow u + C = \frac{1}{2}\ln^2 y$

$\Rightarrow u = \frac{1}{2}\ln^2 y + C$

$\Rightarrow x = \frac{1}{2}\ln y + \frac{C}{\ln y}$

T5 $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^3 + \left(\frac{dy}{dx}\right)$

令 $u = \frac{dy}{dx}$ 则 $\frac{du}{dx} = u^3 + u \Rightarrow \frac{du}{u^3+u} = dx \quad (u \neq 0)$

$$\frac{du}{u(1+u^2)} = dx \Rightarrow \frac{u\,du}{u^2(1+u^2)} = dx$$

$$\Rightarrow \frac{du^2}{u^2(1+u^2)} = 2dx$$

$$\Rightarrow \left(\frac{1}{u^2} - \frac{1}{1+u^2}\right) du^2 = 2dx$$

$$\Rightarrow \ln\left(\frac{u^2}{1+u^2}\right) = 2x + c$$

$$\Rightarrow \frac{u^2}{1+u^2} = Ce^{2x}$$

$$\Rightarrow 1 - \frac{1}{u^2+1} = Ce^{2x}$$

$$\Rightarrow u^2 = \frac{C_1 e^{2x}}{1 - C_1 e^{2x}}$$

$$\Rightarrow u = \sqrt{\frac{C_1 e^{2x}}{1 - C_1 e^{2x}}}$$

$$\Rightarrow dy = \sqrt{C_1}\sqrt{\frac{1}{1 - C_1 e^{2x}}}\, de^x$$

$$\Rightarrow y = \arcsin\left(\sqrt{C_1}e^x\right) + C_2$$

T6 $y''+y=e^x+\cos x$

其特征方程为 $\lambda^2+1=0$，$\lambda=\pm i$

故 $y''+y=0$ 的通解为 $y=C_1\cos x+C_2\sin x$

拆分为 $y''+y=e^x$ ① $y''+y=\cos x$ ②

对于① 显然有特解 $y_1^*=\frac{e^x}{2}$

对于② 设特解为 $y_2^*=x(A\cos x+B\sin x)$

代入得 $(B-A)\sin x+(A+B)\cos x=\cos x$，$A=B=\frac{1}{2}$

$$y_2^*=\frac{x}{2}(\sin x+\cos x)$$

故通解：$y=C_1\cos x+C_2\sin x+\frac{1}{2}e^x+\frac{x}{2}(\sin x+\cos x)$

T7 考虑 $\sum_{n=2}^{\infty}\left(\frac{1}{\sqrt{n-1}}-\frac{1}{\sqrt{n+1}}\right)=\sum_{n=2}^{\infty}\frac{2}{n-1}\sim\frac{2}{n}$，该级数发散

其部分和记为 S_n，则 $n\to\infty$ 时 $S_n\to\infty$

对于原级数，其部分和 $T_{2n}=S_n$，故 $n\to\infty$ 时 $T_{2n}\to\infty$

$T_{2n+1}=T_{2n}+\frac{1}{\sqrt{2n+1}-1}$ $n\to\infty$ 时 $\frac{1}{\sqrt{2n+1}-1}\to 0$，$T_{2n+1}\to\infty$

故 $T_n\to\infty$，原级数发散

T8 $n\to\infty$ $\sin\frac{1}{n}\sim\left(\frac{1}{n}\right)-\frac{1}{6}\left(\frac{1}{n}\right)^3+o\left(\left(\frac{1}{n}\right)^4\right)$

$\frac{1}{n}-\sin\frac{1}{n}\sim\frac{1}{6}\left(\frac{1}{n}\right)^3$

$$\lim_{n\to\infty}\frac{\left(\frac{1}{n}-\sin\frac{1}{n}\right)^\alpha}{\frac{1}{n^{3\alpha}}}=\lim_{n\to\infty}\frac{\frac{1}{6^\alpha n^{3\alpha}}}{\frac{1}{n^{3\alpha}}}=\frac{1}{6^\alpha}$$

① $\alpha>\frac{1}{3}$ 时 $3\alpha>1$，$\sum_{n=1}^{\infty}\frac{1}{n^{3\alpha}}$ 收敛；而 $0<\frac{1}{6^\alpha}<\frac{1}{\sqrt[3]{6}}$ 由比较法，收敛

② $0<\alpha\le\frac{1}{3}$ 时，$3\alpha\le1$，$\sum_{n=1}^{\infty}\frac{1}{n^{3\alpha}}$ 发散，而 $\frac{1}{6^\alpha}\in\left(\frac{1}{\sqrt[3]{6}},+\infty\right)$，故发散

T9 对于 $\sum_{n=1}^{\infty}2^n\cdot\sin\frac{x}{3^n}$

(1) $x\in[-M,M]$ 时，$\left|\sin\frac{x}{3^n}\right|\le\left|\frac{x}{3^n}\right|\le\frac{M}{3^n}$

极限函数 $f(x)=\lim_{n\to\infty}2^n\sin\frac{x}{3^n}=\lim_{n\to\infty}\left(\frac{2}{3}\right)^n\cdot x=0$

故 $|f_n(x)-f(x)|=\left|2^n\cdot\sin\frac{x}{3^n}\right|\le\left(\frac{2}{3}\right)^n\cdot M$，$n\to\infty$ 时 $\left(\frac{2}{3}\right)^nM\to0$ 一致收敛

(2) $x\in(-\infty,+\infty)$ 时，取 $x_n=3^n$，

$\lim_{n\to\infty}|f_n(x_n)-f(x_n)|=|2^n\cdot\sin1|\to\infty$，故不一致收敛

T10 对 $\sum_{n=1}^{\infty}\frac{(2n)!}{(n!)^2}x^{4n}$，令 $u_n=\frac{(2n)!}{(n!)^2}x^{4n}$

$$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right|=\lim_{n\to\infty}\left|\frac{(2n+2)!}{((n+1)!)^2}\cdot\frac{(n!)^2}{(2n)!}x^4\right|=\lim_{n\to\infty}\left|\frac{(2n+2)(2n+1)}{(n+1)^2}x^4\right|$$

$=4|x^4|$ 故当且仅当 $4|x^4|<1$ 收敛，$x\in\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$

收敛半径为 $R=\frac{1}{\sqrt{2}}$

T11 记 $u_n = \frac{x^{4n+1}}{4n+1}$, $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{4n+1}{4n+5}x^4\right| = |x^4|$

故 $|x|<1$ 时 $-1<x<1$ 时 收敛, $x=1$ 时, $\sum_{n=1}^{\infty}\frac{1}{4n+1} > \sum_{n=1}^{\infty}\frac{1}{4n}$ 发散

$x=-1$ 时, $\sum_{n=1}^{\infty}-\frac{1}{4n+1}$ 也发散

故收敛域 $(-1,1)$

记和函数为 $S(x)$. $S'(x) = \left(\sum_{n=1}^{\infty}\frac{x^{4n+1}}{4n+1}\right)' = \sum_{n=1}^{\infty}\left(\frac{x^{4n+1}}{4n+1}\right)'$ $x\in(-1,1)$

$= \sum_{n=1}^{\infty}x^{4n} = \lim_{n\to\infty}x^4\frac{1-(x^4)^n}{1-x^4} = \frac{x^4}{1-x^4}$

△ 又 $S(0)=0$ $S(x) = \int_0^x \frac{1}{1-x^4}-1\,dx$

$= \frac{1}{2}\int_0^x\left(\frac{1}{1+x^2}+\frac{1}{1-x^2}\right)dx - x$

$= \frac{1}{2}\arctan x + \frac{1}{2}\ln\frac{1+x}{1-x} - x$ 符合 $S(0)=0$

T10 $f(x) = \ln(\sqrt{x^2+1}-x)$

$f'(x) = \frac{\frac{x}{\sqrt{x^2+1}}-1}{\sqrt{x^2+1}-x} = \left(\frac{x}{\sqrt{x^2+1}}-1\right)\left(\sqrt{1+x^2}+x\right)$

$= \frac{x^2}{\sqrt{1+x^2}} - \sqrt{1+x^2} = -\frac{1}{\sqrt{1+x^2}}$

$(1+x^2)^{-\frac{1}{2}} = 1+\left(-\frac{1}{2}\right)x^2 + \frac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!}x^4 + \cdots + \frac{\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\cdots\left(-\frac{1}{2}-n+1\right)}{n!}x^n + \cdots$

$= 1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 + \cdots + \frac{(2n-1)!!}{(-2)^n n!}x^n + \cdots$

两边积分 $-\ln(\sqrt{x^2+1}-x) = x - \frac{1}{6}x^3 + \frac{3}{40}x^5 + \cdots + \frac{(2n-1)!!}{(-2)^n(n+1)!}x^{n+1} + \cdots$

$\ln(\sqrt{x^2+1}-x) = -x + \frac{1}{6}x^3 - \frac{3}{40}x^5 + \cdots + \frac{-(2n-1)!!}{(-2)^n(n+1)!}x^{n+1} + \cdots$